Instructions for the Clause 25 Artificial Intelligence (AI)-driven Large Language Model (LLM) Chatbot for Information Retrieval Prototype

CONTENTS

1 Introduction

Tenderers are required to develop a prototype of an AI-driven LLM knowledge management chatbot using the provided mock dataset and sample queries as part of their proposal in the tender submission. Tenderers may include additional materials and queries to enhance the prototype's performance and demonstrate the capabilities of the proposed solution using this prototype. The main purpose is to use the prototype to illustrate the strengths and differentiating capabilities of the proposed solution, demonstrate a credible pathway to scaling the solution for HDB's requirements, and provide evidence of the system's performance, accuracy, and robustness.

Shortlisted tenderers will be invited for an interview, during which the prototype must be fully functional to enable testing of both its capabilities and performance on a hidden test set. The objectives of this prototype are to:

- Evaluate the tenderer's ability to design and implement a knowledge management chatbot at scale
- Examine the tenderer's proficiency in integrating LLMs, employing retrieval mechanisms, and ensuring high response performance.
- Evaluate the performance and robustness of the proposed solution.

2 Problem Statement

The provided mock dataset is related to the Housing, Construction, and Sustainability Authority (HCSA), a government agency in Pulau Ulu. Background information about the mock dataset is included in section 5. The dataset represents a small subset of the documents managed by HCSA.

HCSA's information is stored across multiple sources, including policies, email correspondence, reports, and structured datasets. As a result, officers often need to manually search through different folders and document types to locate the information required for their work. This search process can be time-consuming, potentially taking hours or even days depending on the complexity of the request and the number of documents involved. In some cases, despite significant effort, the required information may not be easily located.

HCSA wants to develop an AI-driven LLM knowledge management chatbot prototype that enables staff to search across policies, email correspondence, reports, and structured datasets in an integrated manner, with the goal of reducing information retrieval time. Tenderers are required to propose and develop a solution that allows staff to efficiently and reliably retrieve relevant information from multiple sources.

3 AI-driven LLM Knowledge Management Chatbot Prototype Requirements

The prototype shall be able to perform the following functions:

- Process natural language queries
- Retrieve relevant information from the provided dataset only
- Provide context-aware responses
- Provide source citations for answers

The responses generated by the prototype do not need to be identical to the expected answers. However, generated responses should be factually correct and not deviate significantly from the expected answer. Responses that are excessively long, contain irrelevant information, or fail to address the question will be considered incorrect, as this indicates the prototype's inability to respond in accordance with user requirements. The set of provided expected answers may contain errors and should not be regarded as conclusive. If the prototype is unable to generate the expected answer, the proposal shall include an analysis of the causes of the incorrect response and strategies for enhancing performance to meet the desired outcomes.

Tenderers may augment the mock dataset with supplementary data to enhance the prototype’s performance and demonstrate the capabilities of their solution. However, the performance of the chatbot will be evaluated on the mock dataset as the primary knowledge base.

The following metrics for the prototype should be included in the proposal. Tenderers may include additional metrics. Please refer to Annex A for details on the performance metrics:

- Accuracy
- Recall rate
- Precision rate
- Completeness rate
- Faithfulness score

The prototype shall showcase the proposed User Interface (UI) and User Experience (UX) design of the AI-drive LLM knowledge management chatbot. Please refer to Annex B for details regarding the requirements for pages that are mandatory or optional, as well as functional or non-functional.

4 Deliverables

The deliverables for the AI-driven LLM knowledge management chatbot prototype will include, but are not limited to, the proposal, the sample query results detailed in Annex C, a private link or screenshots of the prototype, and any supplementary materials. Please refer to Annex D for the description of each column in Annex C. The proposal shall provide detailed information including, but not limited to, the technical approach to developing a high-performing chatbot, data processing techniques, and the performance metrics of the chatbot. This includes the selected Large Language Model (LLM), vector database, embedding models, specific chunking strategy, system guardrails, and data orchestration techniques used to develop a knowledge management solution with high performance, characterized by high accuracy, recall rate, precision rate, completeness rate, and faithfulness. A strong justification for the chosen methodology, in comparison to other alternatives, shall also be provided.

5 Background Information on Mock Dataset

5.1 <u>About Pulau Ulu</u>

Pulau Ulu is a small developing country founded in 1980, with a total land area of 800 square kilometres. It is home to approximately 5,000,000 residents, and its growing population is driving increasing demand for high-quality housing, infrastructure, and public services.

Pulau Ulu is committed to fostering a safe and secure environment with low crime rates, where residents feel comfortable and protected in their daily lives. The country is also dedicated to providing affordable housing, reliable infrastructure, commercial facilities near residential areas, and accessible public amenities for all residents, which support population growth, economic development, and overall social well-being.

At the same time, environmental sustainability is a key priority for Pulau Ulu. With limited land and finite natural resources, the country adopts responsible land-use practices, protects natural Eco system, promotes renewable energy, and implements measures to mitigate climate change. Sustainable development serves as a guiding principle in shaping policies, infrastructure planning, and future growth. Additionally, eco rebates are granted to contractors who fulfil the rebate requirements.

The government established the Housing, Construction, and Sustainability Authority (HCSA) to manage housing development, commercial facilities, infrastructure projects, and sustainability initiatives across the country. Its role is to ensure that development supports long-term environmental, social, and economic needs.

HCSA's key responsibilities include:

- Plan and develop new residential housing projects
- Manage the construction of essential infrastructure such as roads, drainage prototypes, and utilities
- Provide public facilities, including schools, healthcare centres, and community spaces
- Managing commercial properties to support local businesses and economic activity

- Enforce building standards and safety regulations
- Promote sustainability measures in development projects

5.2 Regulatory Framework of HCSA

HCSA develops and implements a comprehensive framework of policies and Standard Operating Procedures (SOPs) to regulate housing development, infrastructure projects, and sustainability initiatives across Pulau Ulu. These policies ensure that all developments are carried out in a consistent, transparent, and accountable manner, in line with established standards and long-term planning objectives.

To implement development projects, HCSA regularly calls for open tenders for residential housing construction, essential infrastructure works such as roads, drainage prototypes, and utilities, as well as public facilities including schools and healthcare centres. All tender processes are conducted in accordance with established procurement guidelines to ensure fairness, transparency, and competitiveness.

Successful tenderers are required to strictly comply with HCSA's regulations, technical specifications, safety standards, and sustainability requirements. Contractors who fail to meet these standards, breach contractual obligations, or demonstrate poor performance may face penalties, termination of contracts, or disqualification from participating in future tenders.

In addition to development oversight, HCSA manages a portfolio of commercial properties. These properties are leased to local businesses for retail shops, office spaces, or other approved commercial uses, and may also be allocated for certain public facilities where appropriate. HCSA oversees tenant management, lease administration, rental collection, and compliance monitoring to ensure proper and lawful use of the premises.

HCSA works closely with both construction partners and commercial tenants, playing a central role in regulating development activities while supporting economic growth and community needs across Pulau Ulu.

5.2.1 Organisational Structure of HCSA

HCSA operates under a comprehensive organizational structure designed to ensure effective leadership and management. Please refer to Annex E for the Organizational Chart of HCSA.

- The Chairperson provides overarching leadership and strategic direction for the organization.
    - The Chief Executive Officer (CEO) is responsible for overall management and operational success, implementing the vision and strategy established by the Chairperson.
        - The Deputy Chief Executive Officer (Building) (DCEO(B)) manages all building-related operations, ensuring that facilities are safe, efficient, and aligned with the organization's goals.
            - The Health, Safety, and Environment (HSE) department implements initiatives related to health, safety, and environmental compliance throughout the organization.
            - The Development Projects department supervises all development initiatives, ensuring that projects are completed on time and within budget.
        - The Deputy Chief Executive Officer (Estate) (DCEO(E)) manages estate operations, ensuring properties are maintained and developed sustainably.
            - The Sustainability and Environment department leads initiatives to integrate sustainability into all organizational practices.
            - The Commercial Properties and Maintenance department manages the administration and maintenance of commercial properties.

5.3 Overview of HCSA Document Organization

The HCSA documents are organized in the folders below:

1	SOPs & Policies Folder	Stores official policies, guidelines, and Standard Operating Procedures (SOPs) that govern housing development, infrastructure projects, commercial facilities management and development, sustainability initiatives, procurement processes, and regulatory enforcement. *Note: HCSA dictates the Contractor's internal SOPs regarding operational procedures and the approvals required within the company. Please refer to Annex F for the recommended organizational chart for the Contractor as provided by HCSA.*
2	Email Repository Folder	Stores daily correspondence, including but not limited to internal communications, clarifications, approvals, and operational discussions between HCSA's officers and external stakeholders.
3	Reports Folder	Stores HDB's financial documentation and annual reports. These reports should be regarded as HCSA's reports for the AI-driven LLM knowledge management chatbot prototype. The prototype should regard HDB and HCSA as a single entity, allowing users to query information using HDB and HCSA interchangeably.
4	Structured Datasets Folder	Stores structured data in a tabular format, including information on development projects, contractors, site inspection results, and permits, which are used for analysis and reporting.

5.3.1 Structured Datasets

Four structured datasets are provided in the mock data. Please refer to the descriptions below for information on each structured dataset:

File Name	Description	Remarks
Contractor listing.xlsx	This file contains details about all past and current contractors that HCSA has engaged.	HCSA can engage multiple contractors, and a single contractor may be involved in several development projects.
Development Projects.xlsx	This file contains information about all HCSA development projects.	One development project is assigned to a single contractor.
Permits.xlsx	This file contains information on permits that have been applied for and approved for development projects.	One development project may require multiple permits.
Inspections.xlsx	This file includes information regarding inspections carried out for the development projects.	One development project can undergo multiple inspections.

Annex A: Computation of performance metrics.

KPI	Formula	Remarks
Accuracy	Accuracy = (Number of correct responses / Total Responses) x 100 Number of correct responses: Total number of factually correct answers provided by the System. Total Responses: Total number of responses generated by the System.	This metric measures the percentage of responses that are correct. If the System answers 8 out of 10 queries factually correctly, the accuracy is 80%.
Recall rate	Recall = (Number of relevant paragraph(s) retrieved / Number of relevant paragraph(s) in the knowledge base) x 100 Number of relevant paragraph(s) that were retrieved: True Positives (TP) Number of relevant paragraph(s) that are not retrieved: False Negatives (FN) Number of relevant paragraph(s) in the knowledge base: True Positives (TP) + False Negatives (FN)	This metric measures the percentage of relevant references that are successfully retrieved. If the System is expected to retrieve 10 paragraphs but only retrieves 5, the recall rate is 50%.
Precision	Precision = (Number of relevant paragraph(s) retrieved / Number of paragraph(s) retrieved) x 100 Number of relevant paragraph(s) retrieved: True Positives (TP) Number of retrieved paragraph(s) that are irrelevant.: False Positives (FP) Number of paragraph(s) retrieved: True Positives (TP) + False Positives (FP)	This metric measures the percentage of relevant references in relation to all references provided by the System. If the System retrieved10 paragraphs and only 5 are relevant, the precision rate is 50%.
Completeness rate	Completeness rate = (Number of key points in the generated response / Number of key points in the expected answer) x 100	This metric reflects the percentage of required information included in the generated response to answer the question. If the expected

KPI	Formula	Remarks
		answer contains 5 key points and the generated response includes only 2, the completeness rate is 40%.
Faithfulness score	Faithfulness Score = Number of claims in the response supported by context / Total number of claims in the response	This metric measures the percentage of claims in the generated response that are supported by the retrieved source evidence. If the generated response contains 5 pieces of factual information and only 2 are supported, the faithfulness score is 40%.

Annex B: Prototype User Interface and User Experience (UI/UX) Requirements

Page Name	Requirement	Functionality[1]
Login page	Mandatory	Functional
User profile page	Optional	Non-Functional
Chat interface page	Mandatory	Functional
Document generation page	Mandatory	Non-Functional
User guide page	Optional	Non-Functional
File upload page	Mandatory	Non-Functional
User management page	Mandatory	Non-Functional
Knowledge base management page	Mandatory	Non-Functional
Conversation history page	Mandatory	Non-Functional
Automated query testing and response evaluation page	Mandatory	Non-Functional
System performance dashboard page	Mandatory	Non-Functional
Chatbot configuration page	Mandatory	Non-Functional
System consumption page	Mandatory	Non-Functional

[1] **Functional**: working features that can be interacted with. **Non-Functional**: static UI/UX mock-ups

Annex C: Sample Query

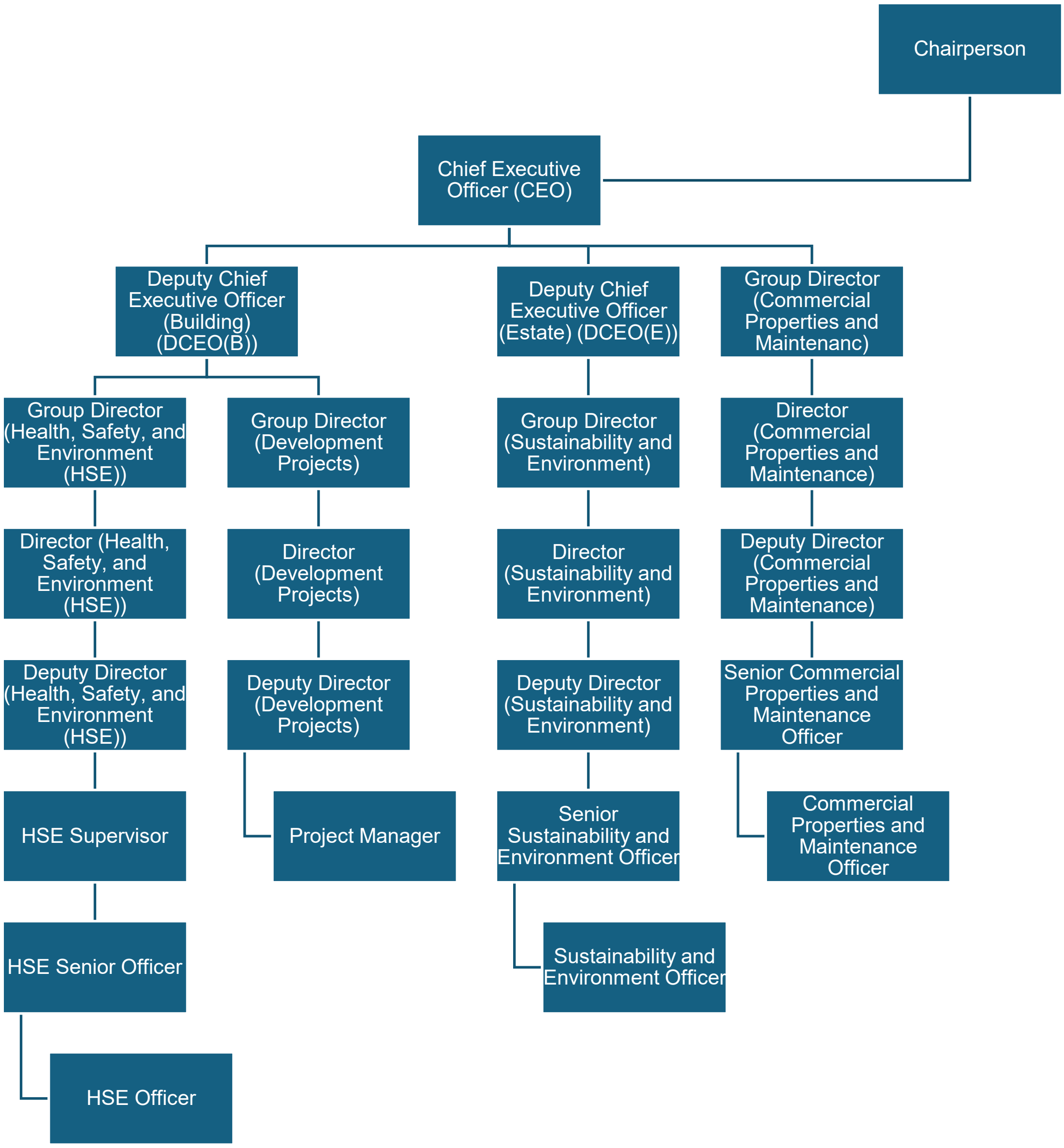
Please refer to the accompanying Excel file "Queries.xlsx".

Annex D: Description of each column in Annex C.

Column	Name of Column	Remarks	Input By
B	Topic of the query	The subject matter of the inquiry.	Provided
C	Sample query	The prompt provided to the prototype to obtain relevant information.	Provided
D	Citation source(s)	The source(s) that the prototype should reference before generating a response.	Provided
E	Relevant paragraph(s) containing supporting evidence	The paragraph(s) that the prototype should reference before generating a response.	Provided
F	Expected answer	This refers to the expected response for the sample query.	Provided
G	Response generated by the chatbot	This refers to the response generated by the prototype.	Tenderer
H	Source(s) referenced by the chatbot	The document from which the prototype extracted the relevant paragraph(s).	Tenderer
I	Paragraph(s) referenced by the chatbot	The relevant paragraph(s) referenced by the prototype to generate the response.	Tenderer
J	Factually correct (Yes / No)	Indicate "Yes" if the response is similar to the expected answer and indicate "No" if it is not.	Tenderer
K	Number of relevant paragraph(s) in the knowledge base	The number of references in the “Relevant paragraph(s) containing supporting evidence” column.	Provided
L	Number of paragraph(s) retrieved	The number of relevant paragraph(s) referenced by the	Tenderer

		prototype for the generated response.	
M	Number of relevant paragraph(s) retrieved	The number of relevant paragraph(s) in the "Number of relevant paragraph(s) retrieved" column.	Tenderer
N	Number of key points in the expected answer	The number of required facts in the expected answer	Provided
O	Number of key points in the generated response	The number of facts in the response generated	Tenderer
P	Number of claims in the response supported by context	This is the number of claims that are supported by information in the retrieved source evidence.	Tenderer
Q	Number of claims in the generated response	The number of claims in the generated response	Tenderer
R	Remarks	Additional notes or supplementary information, if any.	Tenderer

Annex E: Organizational Chart of HCSA.



Annex F: Recommended Organizational Chart for the Contractor by HCSA